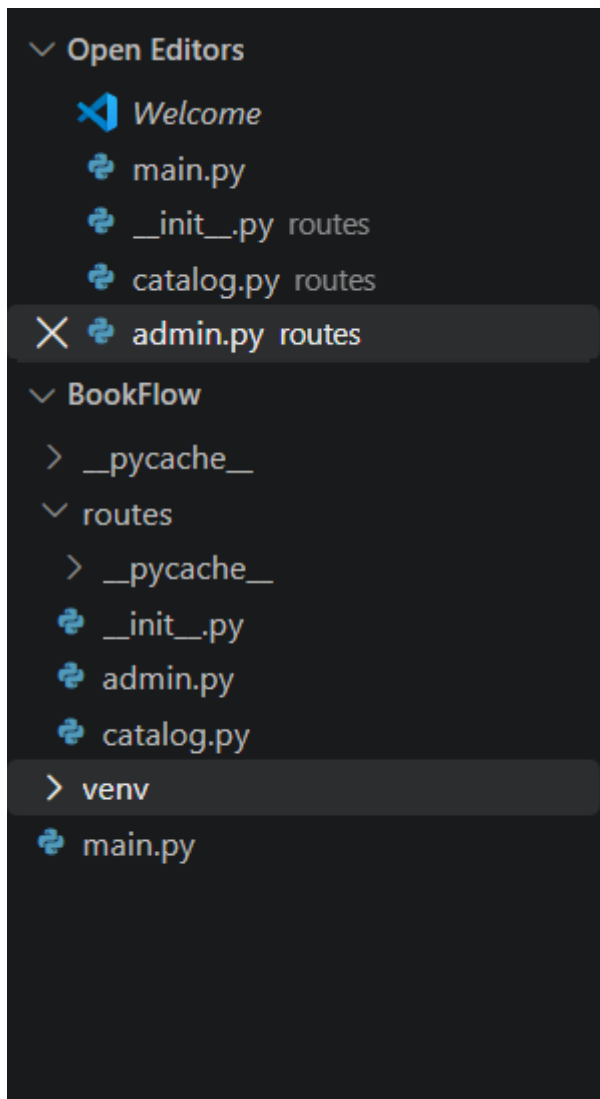


PRACTICAL 6 DBMS



MAIN:

```
1 from fastapi import FastAPI
2 from routes.catalog import router as catalog_router
3 from routes.admin import router as admin_router
4
5 app = FastAPI(
6     title="BookFlow Library API"
7 )
8
9 app.include_router(catalog_router)
10 app.include_router(admin_router)
```

CATALOG:

```
1 from fastapi import APIRouter, HTTPException
2
3 router = APIRouter()
4
5 books = [
6     {
7         "id": 101,
8         "title": "Python Programming",
9         "author": "John Smith",
10        "price": 450,
11        "isbn": "9781234567890",
12        "status": "Available"
13    },
```

```
25 @router.get("/books")
26 def get_books():
27     return books
28
29
30 @router.get("/books/{book_id}")
31 def get_book(book_id: int):
32     for book in books:
33         if book["id"] == book_id:
34             return book
35
36     raise HTTPException(
37         status_code=404,
```

```
37         status_code=404,
38         detail="Book not found"
39     )
```

ADMIN:

```
1 from fastapi import APIRouter, HTTPException
2 from pydantic import BaseModel, Field
3 from routes.catalog import books
4
5 router = APIRouter()
6
7
8 class Book(BaseModel):
9     title: str
10    author: str
11    price: float = Field(ge=0)
12    isbn: str
13    status: str
```

```
16 @router.post("/admin/books")
17 def add_book(book: Book):
18     new_id = max([b["id"] for b in books], default=100) + 1
19
20     new_book = {
21         "id": new_id,
22         **book.model_dump()
23     }
24
25     books.append(new_book)
26
27     return {
28         "message": "Book added successfully",
29         "book": new_book
```

```

33 @router.put("/admin/books/{book_id}")
34 def update_book(book_id: int, book: Book):
35     for index, existing_book in enumerate(books):
36         if existing_book["id"] == book_id:
37
38             updated_book = {
39                 "id": book_id,
40                 **book.model_dump()
41             }

```

```

43         books[index] = updated_book
44
45     return {
46         "message": "Book updated successfully",
47         "book": updated_book
48     }
49
50     raise HTTPException(
51         status_code=404,
52         detail="Book not found"
53     )

```

BookFlow Library API 0.1.0 OAS 3.1

/openapi.json

default

GET	/books	Get Books	⌵
GET	/books/{book_id}	Get Book	⌵
POST	/admin/books	Add Book	⌵
PUT	/admin/books/{book_id}	Update Book	⌵
DELETE	/admin/books/{book_id}	Delete Book	⌵

Schemas

Book > Expand all object

HTTPValidationError > Expand all object

ValidationError > Expand all object

GET

/books

Get Books

⌵

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

200

Response body

```
{
  "id": 101,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 450,
  "isbn": "9781234567890",
  "status": "Available"
},
{
  "id": 102,
  "title": "Database Management",
  "author": "Robert Brown",
  "price": 550,
  "isbn": "9780876543210",
  "status": "Available"
}
]
```

Download

Response headers

```
content-length: 240
content-type: application/json
date: Tue, 15 Sep 2026 06:11:14 GMT
server: uvicorn
```

GET

/books/{book_id}

Get Book

⌵

Parameters

Cancel

Name

Description

book_id * required

integer

(path)

101

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books/101' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/books/101

Server response

Code

Details

200

Response body

```
{
  "id": 101,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 450,
  "isbn": "9781234567890",
  "status": "Available"
}
```

Download

Response headers

```
content-length: 117
content-type: application/json
date: Tue, 15 Sep 2026 06:12:23 GMT
server: uvicorn
```

Responses

Code

Description

Links

POST

/admin/books Add Book

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{  "title": "Web Development",  "author": "David Lee",  "price": 600,  "isbn": "9781111111111",  "status": "Available"}  
```

Curl

```
curl -X 'POST' \  'http://127.0.0.1:8000/admin/books' \  -H 'accept: application/json' \  -H 'Content-Type: application/json' \  -d '{  "title": "Web Development",  "author": "David Lee",  "price": 600,  "isbn": "9781111111111",  "status": "Available"}'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code

Details

200

Response body

```
{  "message": "Book added successfully",  "book": {    "id": 103,    "title": "Web Development",    "author": "David Lee",    "price": 600,    "isbn": "9781111111111",    "status": "Available"  }  }
```

Response headers

```
content-length: 160  content-type: application/json  date: Tue, 15 Sep 2026 06:33:15 GMT  server: uvicorn
```

Responses

PUT

/admin/books/{book_id} Update Book

Parameters

Name	Description
book_id required	103
integer	
(path)	

Request body required

application/json

Edit Value | Schema

```
{  "title": "Advanced Web Development",  "author": "David Lee",  "price": 750,  "isbn": "9781111111111",  "status": "Available"}  
```

Execute

Clear

Responses

Curl

```
curl -X 'PUT' \
'http://127.0.0.1:8000/admin/books/103' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "title": "Advanced Web Development",
  "author": "David Lee",
  "price": 750,
  "isbn": "9781111111111",
  "status": "Available"
}'
```

Request URL

```
http://127.0.0.1:8000/admin/books/103
```

Server response

Code

Details

200

Response body

```
{
  "message": "Book updated successfully",
  "book": {
    "id": 103,
    "title": "Advanced Web Development",
    "author": "David Lee",
    "price": 750,
    "isbn": "9781111111111",
    "status": "Available"
  }
}
```

Download

Response headers

```
content-length: 171
content-type: application/json
date: Tue, 15 Sep 2026 06:34:37 GMT
server: uvicorn
```

Responses

Code

Description

Links

DELETE /admin/books/{book_id} Delete Book

Cancel

Parameters

Name	Description
book_id * required	
integer	103
(path)	

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
'http://127.0.0.1:8000/admin/books/103' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/admin/books/103
```

Server response

Code

Details

200

Response body

```
{
  "message": "Book deleted successfully",
  "book": {
    "id": 103,
    "title": "Advanced Web Development",
    "author": "David Lee",
    "price": 750,
    "isbn": "9781111111111",
    "status": "Available"
  }
}
```

Download

Response headers

```
content-length: 171
content-type: application/json
date: Tue, 15 Sep 2026 06:35:50 GMT
server: uvicorn
```

Responses